FLEET TEAM DIGEST

737MAX-FTD-28-18002

Issue Title : Fuel Quantity Indication System Accuracy

Airplane Model	ATA	Minor Model(s)
737MAX	2841-00	-7, -8, -8200, -9, -BBJ

Other Model(s)	ECCN
757, 787, 777, 737, 737NG, 747	9E991

Originated Date	Last Revised Date	Created On
11/07/2018	11/07/2018	11/02/2018

Estimated Completion Date	Next Update Date

Applicability

All 737-8/-8200/-9/BBJ aircraft.

Description

This FTD article gives the accuracy of the FQIS system and the accuracy of the fuel flow transmitter.

Background

On 16 April 2009, the European Union adopted Decision 2009/339/EC. This Decision amends 2008/589/EC to include aviation activities in the monitoring and reporting guidelines for emissions. Directive 2008/101/EC amended directive 2003/87/EC to include aviation activities in the scheme for greenhouse gas emissions allowance trading. Virtually all airlines with operations to, from and within the EU fall under the scope of the directives, including non-EU airlines.

According to IATA, this EU directive will require airlines that fly to or within Europe to:

- Submit monitoring plans by 31 August 2009

- Monitor ton-kilometers and CO2 emissions from 1 January 2010

- Report ton-kilometer data by 31 March 2011

- Report CO2 emissions data by 31 March 2011

- Apply for free emissions allowances by 31 March 2011

- Surrender allowances for 2012 emissions by 30 April 2013

Decision 2009/339/EC adds "Annex XIV - Activity-specific guidelines for aviation activities as listed in Annex I to Directive 2003/87/EC". This annex provides calculation formulas for determining fuel consumption, "emission factor", and requires airlines to understand the "uncertainty assessment." Part of this "uncertainty assessment" is the accuracy of the fuel quantity indication system and the accuracy of the engine fuel flow transmitter.

The following are links to this directive and decision:

BOEING PROPRIETARY - The information contained herein is Proprietary to The Boeing Company and shall not be reproduced or disclosed in whole or in part or used for any reason except when such user possesses direct, written authorization from The Boeing Company. Encrypted emails required for forwarding via Email.The document generated on 11/05/2026 by Fleet Team Digest application via MyBoeingFleet. This document is for reference purposes only, visit the application for all the latest and updated revisions of this document.

Copyright ©2026 Boeing. All rights reserved.

EXPORT CONTROLLED -This technology or software is subject to the U.S. Export Administration Regulations (EAR), (15 C.F.R. Parts 730-774). No authorization from the U.S. Department of Commerce is required for export, re-export, in-country transfer, or access EXCEPT to country group E:1 or E:2 countries/persons per Supp.1 to Part 740 of the EAR. ECCN : 9E991

FLEET TEAM DIGEST

Decision 2009/339/EC: http://eur-lex.europa.eu/LexUriServ/LexUriServ.do?uri=OJ:L:2009:103:0010:0029:EN:PDF

Directive 2008/101/EC: http://eur-lex.europa.eu/LexUriServ/LexUriServ.do?uri=OJ:L:2009:008:0003:0021:EN:PDF

This FTD article provides airlines with the accuracy of the fuel quantity and fuel used indications to support the European Union reporting requirements.

Status

The 737Max fuel quantity indication system has the following accuracy tolerance:

737Max with densitometer, FQIS accuracy:

+/- 1.0% overall of each tank on the ground.

Mains and Center tank each > 50% full: +/- 1.5% where -1 to 5 deg pitch, +/- 1 deg roll in steady in flight.

Mains and Center tank each < 50% full: +/- 1.0% where -1 to 5 deg pitch, +/- 1 deg roll in steady in flight.

737Max without densitometer, FQIS accuracy:

+/- 2.0% overall of each tank on the ground.

Mains and Center tank each > 50% full: +/- 2.5% where -1 to 5 deg pitch, +/- 1 deg roll in steady in flight.

Mains and Center tank each < 50% full: +/- 2% where -1 to 5 deg pitch, +/- 1 deg roll in steady in flight.

The total tolerance for the FQIS system is based on a full tank of each tank. For example, if the main fuel tank maximum capacity is 4147 KG based on specific gravity of .8507 KG/L, and when the tank fuel level is less than 50% full, then the system tolerance of the gauging is 0.02 (airplane without a densitometer) * 4147 = +/- 83 KG in steady in flight.

When the tank is more than 50% full, the system tolerance is 0.025 * 4147 KG = +/- 104 KG in steady in flight.

When on the ground, the system tolerance is 0.02 * 4147 KG = +/- 83 KG at any fuel level.

These numbers meet the FAA Technical Standard Order TSO-C55a, which gives manufactures the minimum performance standards for approval and identification with the applicable TSO marking. MPD 28-080-00, Operational (BITE) test of the FQIS system, along with flight deck effects, ensure that the system is within its tolerances.

The accuracy tolerance of the fuel flow transmitter is a function of the fuel flow. At engine idle, the system tolerance can be 12%. During cruise, the tolerance is less than 0.5%. The fuel flow indication is integrated over time to calculate the fuel used for each engine.

Interim Action

None. This data is provided to support airlines in conforming to the EU emissions directives.

Final Action

BOEING PROPRIETARY - The information contained herein is Proprietary to The Boeing Company and shall not be reproduced or disclosed in whole or in part or used for any reason except when such user possesses direct, written authorization from The Boeing Company. Encrypted emails required for forwarding via Email.The document generated on 11/05/2026 by Fleet Team Digest application via MyBoeingFleet. This document is for reference purposes only, visit the application for all the latest and updated revisions of this document.

Copyright ©2026 Boeing. All rights reserved.

EXPORT CONTROLLED -This technology or software is subject to the U.S. Export Administration Regulations (EAR), (15 C.F.R. Parts 730-774). No authorization from the U.S. Department of Commerce is required for export, re-export, in-country transfer, or access EXCEPT to country group E:1 or E:2 countries/persons per Supp.1 to Part 740 of the EAR. ECCN : 9E991

FLEET TEAM DIGEST

None. This data is provided to support airlines in conforming to the EU emissions directives.

Operator Action

None. This data is provided to support airlines in conforming to the EU emissions directives.

Reference Categories

Reference Type	Reference Number	Revision Number	Reference Date
Multi-Operator Message (MOM)	MOM-MOM-09-0490-01B		08/20/2009

Related Categories

- Regulatory : Issues mandated by airworthiness directive or other regulatory guidance material

BOEING PROPRIETARY - The information contained herein is Proprietary to The Boeing Company and shall not be reproduced or disclosed in whole or in part or used for any reason except when such user possesses direct, written authorization from The Boeing Company. Encrypted emails required for forwarding via Email.The document generated on 11/05/2026 by Fleet Team Digest application via MyBoeingFleet. This document is for reference purposes only, visit the application for all the latest and updated revisions of this document.

Copyright ©2026 Boeing. All rights reserved.

EXPORT CONTROLLED -This technology or software is subject to the U.S. Export Administration Regulations (EAR), (15 C.F.R. Parts 730-774). No authorization from the U.S. Department of Commerce is required for export, re-export, in-country transfer, or access EXCEPT to country group E:1 or E:2 countries/persons per Supp.1 to Part 740 of the EAR. ECCN : 9E991